

Mohd Farrakh Nizam

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EDUCATION

Bachelor of Technology in AI and ML (CGPA: 8.3/10)

Aug 2023 – Present

Chandigarh Engineering College (CEC), Jhanjeri, Mohali

Senior Secondary (Class XII) (CGPA: 7.0/10)

2022

Sant Atulanand Convent School, Varanasi

TECHNICAL SKILLS

Programming: Python, Java, JavaScript

AI & Machine Learning: Machine Learning (Basics), Generative AI, Agentic AI, LLM Concepts, AI Prompting, PyTorch, Scikit-Learn

AI Tools & Frameworks: React, Express.js, Node.js, LangChain, OpenAI Whisper

Backend & APIs: FastAPI, REST APIs, API Integration, Postman

Data & Databases: SQL, MySQL, MongoDB, Postgres, Pandas, NumPy

Developer Tools: Git, GitHub, Docker, Kubernetes

Cloud Tools: Google Cloud Platform (GCP), IBM Cloud, AWS (Basic), Clustering

INTERNSHIP EXPERIENCE

Research and Development Intern

On-site

National Institute of Technology (NIT) Delhi | [Certificate](#)

June 2026 – Aug 2026

- Conducted research on underwater image improved using GAN and Transformer-based deep learning models to address challenges such as blurriness, color distortion, low contrast, and light scattering in underwater images.
- Enhanced model performance by improving the Structural Similarity Index (SSIM) from **0.81** to **0.91** and increasing the Peak Signal-to-Noise Ratio (PSNR) from **26 dB** to **29 dB** through architecture optimization and hyperparameter tuning.
- Applied, trained, and evaluated deep learning models using **Python, PyTorch, NumPy, Matplotlib**, and image processing techniques to support research experimentation and performance analysis.

Artificial Intelligence Intern

Remote

Infosys Springboard 6.0 | [Certificate](#)

Jan 2024 – Mar 2024

- Optimized audio preprocessing workflows using Librosa by 90%, significantly increasing the throughput for high-volume data extraction tasks.
- Collaborated on responsible AI frameworks to ensure 100% compliance with ethical data handling and gained accuracy on the project above 85% with standards in real-world application prototypes.

PROJECTS

Underwater Image Enhancement using GAN | GAN, CBAM, CNN | [Report PDF](#) | [Certificate](#)

- * Researched and optimized a FUnIE-GAN framework for underwater image enhancement, addressing color distortion, low contrast, haze, and visibility degradation across EUVP and RUIE datasets.
- * Integrated CBAM attention into the Generator and optimized training with PatchGAN, L1, SSIM, and VGG-19 losses, improving **SSIM from 0.81 to 0.91** and **PSNR from 26 to 29 dB**.
- * **Result:** Improved structural detail recovery and image quality over the baseline.
- * **Skills:** Deep Learning, Model Optimization, Research, Analytical Problem-Solving, Team work, Time management

RAG-Based AI Teaching Assistant | [GitHub](#)

- * Architected an end-to-end RAG pipeline using Python, OpenAI Whisper, MoviePy, and ChromaDB to extract, transcribe, and index educational video lectures into persistent vector storage.
- * Optimized character chunking and top-*k* semantic retrieval, improving **retrieval precision by 35%** and mapping responses to precise **MM:SS timestamps**.
- * **Result:** Produced source-grounded answers with strict retrieval constraints and fallback handling.
- * **Skills:** RAG, LLM Integration, Semantic Search, Prompt Engineering

AI Podcast Transcription System | [GitHub](#) | [Live Demo](#)

- * Engineered an automated audio transcription pipeline using OpenAI Whisper, Librosa, and Pandas for processing diverse audio inputs and extracting transcription-related features.
- * Optimized audio preprocessing to reduce **processing latency by 40%** for files up to **50 MB** and enabled remote URL processing through a Streamlit dashboard.
- * **Result:** Enabled remote audio processing up to **5× faster** than manual local uploads.
- * **Skills:** Audio Processing, Python, Data Handling, Streamlit, Team work, Time management

CERTIFICATIONS

AWS Cloud Academy – Data Engineering [View Certificate](#)

Machine Learning and Deep Learning (IBM SkillsBuild) [View Certificate](#)

Deep Learning for Developers (Infosys Springboard) [View Certificate](#)

Introduction to OpenAI GPT Models (Infosys Springboard) [View Certificate](#)

ServiceNow University Internship program [View Certificate](#)

ACHIEVEMENTS / PARTICIPATIONS

Top 50 National Finalist – Hack2Skill Hackathon by Google Cloud [View Certificate](#)

Google Student Ambassador (2025) – Organized and coordinated 5+ technical workshops, strengthening communication, presentation, and student engagement skills.

Team Lead – College Student Community – Led and coordinated team activities, delegated tasks, and collaborated with members to achieve community objectives.